

自动控制原理知识点很多，但出题相对固定；控制理论的诸多不完整、不严谨的使用与方法姑且用着，以后有需要再作考量；801控制原理的考察范围不完全明晰，视为风险的一部分；总结必要的概念、结论、方法、经验、典例等，仅为知识体系，不覆盖做题分析能力（灵活使用，掌握足够的技巧方法，面对题目“非无之无”）考试面前，适当理解以对抗风险，但应用为先！

默认章节

考情分析

考察点主要在控制理论，电路，电机，模电等知识点涉及不深

常见函数拉氏变换

$$\delta(t), \delta(t - \tau_0), 1(t), t, \frac{t^n}{n!}, e^{\pm at}, \sin \omega t, \cos \omega t$$

$$\begin{array}{cc} f(t) & F(s) \\ \delta(t) & 1 \\ \delta(t - \tau_0) & e^{-\tau_0 s} \end{array}$$

$$\begin{array}{cc} 1(t) & \frac{1}{s} \end{array}$$

$$\begin{array}{cc} t & \frac{1}{s^2} \end{array}$$

$$\begin{array}{cc} \frac{t^n}{n!} & \frac{1}{s^{n+1}} \end{array}$$

$$\begin{array}{cc} e^{\pm at} & \frac{1}{s \mp a} \end{array}$$

$$\begin{array}{cc} \sin \omega t & \frac{\omega}{s^2 + \omega^2} \end{array}$$

$$\begin{array}{cc} \cos \omega t & \frac{s}{s^2 + \omega^2} \end{array}$$

留数法(单根，重根)

$$\begin{array}{l} \text{单根: } \frac{c_i}{s - p_i} \\ c_i = \lim_{s \rightarrow p_i} [(s - p_i)F(s)] \end{array}$$

$$\begin{array}{l} \text{重根: } \frac{c_m}{(s - p_1)^m}, \frac{c_{m-1}}{(s - p_1)^{m-1}}, \dots \\ c_m = \lim_{s \rightarrow p_1} [(s - p_1)^m F(s)] \end{array}$$

$$c_{m-1} = \frac{1}{1!} \cdot \lim_{s \rightarrow p_1} \frac{d}{ds} [(s - p_1)^m F(s)]$$

$$\begin{array}{l} c_{m-2} = \frac{1}{2!} \cdot \lim_{s \rightarrow p_1} \frac{d^2}{ds^2} [(s - p_1)^m F(s)] \\ \dots \end{array}$$

拉氏变换基本定理

线性性质、微分定理、积分定理、实位移定理、复位移定理、初值定理、终值定理

$$\mathcal{L}[af_1(t) \pm bf_2(t)] = aF_1(s) \pm bF_2(s)$$

$$\begin{cases} \mathcal{L}[f^{(n)}(t)] = s^n F(s) \quad (\text{零初始条件}) & \mathcal{L}[f'(t)] = sF(s) - f(0) \\ \mathcal{L}[f^{(n)}(t)] = s^n F(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots - sf^{(n-2)}(0) - f^{(n-1)}(0) \end{cases}$$

$$\begin{cases} \mathcal{L}\left[\underbrace{\int \cdots \int}_n f(t)(dt)^n\right] = \frac{1}{s^n} F(s) \quad (\text{零初始条件}) & \mathcal{L}[\int f(t)dt] = \frac{1}{s} F(s) + \frac{1}{s} f^{(-1)}(0) \\ \mathcal{L}\left[\underbrace{\int \cdots \int}_n f(t)(dt)^n\right] = \frac{1}{s^n} F(s) + \frac{1}{s^n} f^{(-1)}(0) + \frac{1}{s^{n-1}} f^{(-2)}(0) + \cdots + \frac{1}{s} f^{(-n)}(0) \end{cases}$$

$$\mathcal{L}[f(t \pm \tau_0)] = e^{\pm \tau_0 s} F(s) \quad \text{注: } \mathcal{L}[f'(t \pm \tau_0)] \xrightarrow[g(t)=f(t \pm \tau_0)]{g(t)=f'(t)} e^{\pm \tau_0 s} sF(s)$$

$$\mathcal{L}[e^{\mp At} f(t)] = F(s \pm A) \quad \text{注: } \mathcal{L}[e^{\mp At} f'(t)] \xrightarrow[g(t)=f'(t)]{g(t)=f(t)} G(s \pm A) = (s \pm A)F(s \pm A)$$

$$\lim_{t \rightarrow 0} f(t) = \lim_{s \rightarrow \infty} s \cdot F(s)$$

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} s \cdot F(s)$$